Python Dictionaries Master Test

30 Multiple Choice Questions (Theoretical Concepts & Practical Code Problems)

Subject: Python Programming **Topic:** Dictionaries (dict) **Total Questions:** 30 **Format:** MCQ with Answer Key

Q1. How do you create an empty dictionary in Python?

A) `my_dict = {}`
B) `my_dict = dict()`
C) Both A and B
D) `my_dict = []`

Q2. What will be the output of the following code?

```
d = {'a': 1, 'b': 2, 'a': 3}
print(d['a'])
```

A) `1`
B) `3`
C) `{'a': 1, 'b': 2, 'a': 3}`
D) `KeyError`

Q3. Which of the following data types can serve as a valid dictionary key in Python?

A) `list` (`[1, 2]`)
B) `dict` (`{'x': 1}`)
C) `tuple` (`(1, 2)`)
D) `set` (`{1, 2}`)

Q4. What will happen if you try to access a missing key using bracket notation (e.g., `d['missing']`)?

A) It returns `None`
B) It returns `False`
C) It raises a `KeyError`
D) It creates the key with value `0`

Q5. Which dictionary method safely retrieves a value for a key without raising an error if the key is missing?

A) `d.fetch(key)`
B) `d.get(key)`
C) `d.find(key)`
D) `d.retrieve(key)`

Q6. What will be the output of this code?

```
d = {1: 'one', 2: 'two', 3: 'three'}
print(2 in d)
```

A) `True`
B) `False`
C) `'two'`
D) `Error`

Q7. How do you add a new key-value pair or update an existing key in a dictionary?

A) `d.add(key, value)`

B) `d[key] = value`

C) `d.insert(key, value)`

D) `d.append(key, value)`

Q8. What does the `pop()` method do when called on a dictionary with a specific key (e.g., `d.pop('k')`)?

A) Removes key and returns its value.

B) Returns last inserted pair as tuple.

C) Deletes entire dictionary from memory.

D) Clears all key-value pairs.

Q9. Starting with Python 3.7+, what is a guaranteed characteristic of dictionary keys?

A) Automatically sorted alphabetically/numerically.

B) Maintain insertion order.

C) Stored in randomized order.

D) Must be strictly immutable strings.

Q10. What will be the output of the following code?

```
d = {'x': 10, 'y': 20}
print(type(d.keys()))
```

A) `<class 'list'>`

B) `<class 'tuple'>`

C) `<class 'dict_keys'>`

D) `<class 'set'>`

Q11. Which method removes and returns the *last inserted* key-value pair from a dictionary as a 2-tuple?

A) `pop()`

B) `popitem()`

C) `remove()`

D) `clear()`

Q12. What will be the output of this code snippet?

```
student = {'name': 'Alice', 'age': 25}
student.update({'age': 26, 'city': 'New York'})
print(student['age'])
```

A) `25`

B) `26`

C) `'New York'`

D) `KeyError`

Q13. What is the average time complexity for key lookup, insertion, and deletion in a Python dictionary?

A) O(n) - Linear time

B) O(log n) - Logarithmic time

C) O(1) - Constant time

D) $O(n^2)$ - Quadratic time

Q14. What will happen if you pass a list of tuples like `[('a', 1), ('b', 2)]` into `dict()`?

A) Raises a `TypeError`.

B) Creates dictionary `{'a': 1, 'b': 2}`.

C) Creates a nested list of dictionaries.

D) Returns a tuple of keys.

Q15. What does `setdefault()` do if the key *already exists* in the dictionary?

A) Overwrites it with the new default value.

B) Raises a `KeyError`.

C) Returns current value and leaves it unchanged.

D) Returns `None`.

Q16. What will be the output of this code?

```
d1 = {'a': 1, 'b': 2}
d2 = {'b': 99, 'c': 3}
merged = {**d1, **d2}
print(merged['b'])
```

A) `2`

B) `99`

C) `{'b': 2}`

D) `Error`

Q17. How can you remove all key-value pairs from a dictionary at once, leaving it empty (`{}`)?

A) `d.delete()`

B) `d.clear()`

C) `del d`

D) `d.empty()`

Q18. What will be the output of the following code?

```
d = {'a': 1, 'b': 2}
print(len(d))
```

A) `2`

B) `4`

C) `1`

D) `Error`

Q19. Which underlying data structure is used to implement Python dictionaries under the hood?

A) Doubly linked lists

B) Binary search trees

C) Hash tables (hash maps)

D) Dynamic arrays

Q20. What will be the output of this snippet?

```
d = {'a': 10, 'b': 20, 'c': 30}
print(list(d.values()))
```

A) `['a', 'b', 'c']`

B) `[10, 20, 30]`

C) `[('a', 10), ('b', 20), ('c', 30)]`

D) `Error`

Q21. What does the `fromkeys()` class method do?

A) Creates dictionary with keys from iterable and a shared value.

B) Deletes multiple keys simultaneously.

C) Extracts unique keys from a list of dicts.

D) Checks if multiple keys exist.

Q22. What will be the output of the following code?

```
d = {'a': 1, 'b': 2}
print(d.get('c', 42))
```

A) `None`

B) `42`

C) `KeyError`

D) `0`

Q23. Can dictionary values be mutable objects (such as lists or other dictionaries)?

A) No, values must be immutable.

B) Yes, values can be of any data type.

C) Yes, but only if wrapped in a tuple.

D) No, Python raises a `ValueError`.

Q24. What will be the output using the union operator (`|`) in Python 3.9+?

```
d1 = {'apple': 1}
d2 = {'banana': 2}
result = d1 | d2
print(result['banana'])
```

A) `1`

B) `2`

C) `KeyError`

D) `TypeError`

Q25. What happens when you use a `for` loop directly on a dictionary (e.g., `for k in my_dict:`)?

A) Iterates over the values.

B) Iterates over the keys.

C) Iterates over key-value tuples.

D) Raises a `TypeError`.

Q26. What will be the output of this dictionary comprehension?

```
res = {x: x**2 for x in range(3)}
print(res[2])
```

A) `2`

B) `4`

C) `1`

D) `KeyError`

Q27. Why must dictionary keys be hashable?

A) Allows computing index via hash function for fast lookups.

B) Ensures keys are always strings.

C) Prevents dictionaries exceeding memory.

D) Automatically encrypts data for security.

Q28. What will be the output of the following code?

```
d = {'a': 1, 'b': 2}
del d['a']
print(d)
```

A) `{'a': 1, 'b': 2}`

B) `{'b': 2}`

C) `{}`

D) `KeyError`

Q29. What is the correct way to iterate through both keys and values simultaneously?

A) `for k, v in d.items():`

B) `for k, v in d.keys():`

C) `for k, v in d.values():`

D) `for k, v in d:`

Q30. What will be the result of executing `dict(a=1, b=2)`?

A) `{'a': 1, 'b': 2}`

B) `[('a', 1), ('b', 2)]`

C) `('a', 'b')`

D) `TypeError`

Answer Key & Detailed Explanations

Q1. Correct Answer: C

Both `my_dict = {}` and `my_dict = dict()` create an empty dictionary object in Python. Note that `{}` creates a dictionary, not a set.

Q2. Correct Answer: B

Dictionary keys must be unique. When a duplicate key is defined during initialization, the later value overwrites the earlier one, so `'a'` evaluates to `3`.

Q3. Correct Answer: C

Dictionary keys must be hashable (immutable). Tuples are immutable and hashable, whereas lists, dictionaries, and sets are mutable and unhashable.

Q4. Correct Answer: C

Accessing a non-existent key directly via bracket notation `d['missing']` raises a `KeyError` at runtime.

Q5. Correct Answer: B

The `get(key)` method safely retrieves a value without crashing, returning `None` (or a specified fallback value) if the key is missing.

Q6. Correct Answer: A

The `in` membership operator checks for key presence in a dictionary. Since `2` is a valid key in `d`, it returns `True`.

Q7. Correct Answer: B

Assignment syntax `d[key] = value` sets or updates a key-value pair in a dictionary.

Q8. Correct Answer: A

`d.pop(key)` removes the key from the dictionary and returns its corresponding value.

Q9. Correct Answer: B

Starting in Python 3.7+, dictionary insertion order is guaranteed as an official standard language feature.

Q10. Correct Answer: C

The `d.keys()` method returns a dynamic view object of class `<class 'dict_keys'>`.

Q11. Correct Answer: B

`popitem()` removes and returns the last inserted key-value pair as a 2-element tuple `(key, value)`.

Q12. Correct Answer: B

The `update()` method updates overlapping keys in-place (updating `'age'` to `26`) and appends new ones (`'city'`).

Q13. Correct Answer: C

Dictionaries use optimized hash tables, granting O(1) constant average time complexity for lookups, insertions, and deletions.

Q14. Correct Answer: B

Passing an iterable of 2-tuples to the `dict()` constructor converts each pair into a key-value item: `{'a': 1, 'b': 2}`.

Q15. Correct Answer: C

If a key already exists, `setdefault()` returns its existing value and leaves the dictionary unchanged.

Q16. Correct Answer: B

Dictionary unpacking `{**d1, **d2}` merges dictionaries. Overlapping key values from the right-hand dictionary (`d2`) overwrite those in `d1`, setting `'b'` to `99`.

Q17. Correct Answer: B

The `clear()` method removes all key-value entries from the dictionary in place, leaving it as `{}`.

Q18. Correct Answer: A

The `len()` function counts top-level key-value pairs in a dictionary, returning `2`.

Q19. Correct Answer: C

Python dictionaries are implemented using dynamic hash tables (hash maps) under the hood.

Q20. Correct Answer: B

`d.values()` produces a view of all values in the dictionary. Casting it to `list()` yields `[10, 20, 30]`.

Q21. Correct Answer: A

`dict.fromkeys(iterable, value)` constructs a new dictionary with keys drawn from the iterable, setting each key's value to the specified default parameter.

Q22. Correct Answer: B

Since key `'c'` is absent in `d`, `d.get('c', 42)` returns the provided fallback default value `42`.

Q23. Correct Answer: B

While dictionary keys must be hashable/immutable, dictionary values can be of any arbitrary data type (lists, sets, dicts, functions, etc.).

Q24. Correct Answer: B

The `|` union operator introduced in Python 3.9 merges two dictionaries into a new dict. Accessing `result['banana']` yields `2`.

Q25. Correct Answer: B

Iterating over a dictionary directly in a `for` loop iterates through its keys by default.

Q26. Correct Answer: B

The dictionary comprehension builds key-value pairs `x: x**2` for `range(3)` (0, 1, 2). For key `2`, the value is `2**2 = 4`.

Q27. Correct Answer: A

Hashability ensures that a key's calculated hash value remains invariant, enabling instant O(1) memory bucket lookup.

Q28. Correct Answer: B

`del d['a']` deletes the key `'a'` and its value, leaving dictionary `{'b': 2}`.

Q29. Correct Answer: A

The `.items()` method returns key-value tuples, enabling unpacking in a loop: `for k, v in d.items():`.

Q30. Correct Answer: A

The `dict()` constructor accepts keyword arguments (`a=1, b=2`) and constructs the dictionary `{'a': 1, 'b': 2}`.